PAPER A | RESEARCH FRAMEWORK | OCTOBER 2026

When Validation Misleads

Group Dependence, Temporal Change and the Reliability of Performance Estimates in Supply Chain Machine Learning

Tewogbade Shakir Adeyemi

1. Positioning

Supply chain prediction models are usually validated by randomly splitting historical records. Supply chain data seldom suit this: line items share orders, customers return, and operations change over time through new markets, peak seasons, disruptions and revised service promises. Leakage and unsuitable validation are well documented in machine-learning-based research (Kaufman et al., 2012; Kapoor & Narayanan, 2023). Validation designs for dependent and time-ordered data are well studied in other fields (Roberts et al., 2017; Bergmeir & Benítez, 2012; Tashman, 2000). Monitoring of deployed supply chain models has shown that drift can occur without deterioration (Eck et al., 2022).

What is not yet established is how group dependence and temporal change, separately and together, affect performance estimates for supply chain models; whether the usual comparisons between validation designs identify these mechanisms or only describe differences; and when these effects change which model a practitioner would choose. Paper A addresses these questions. It makes no claim to be the first study of leakage or temporal validation in this domain, and a structured literature search will confirm the gap before submission.

2. Aim

To determine how group dependence and temporal change affect estimated predictive performance, calibration and model rankings in supply chain machine learning, and to separate descriptive differences between validation designs from mechanisms established through controlled experiments, using DataCo and Olist.

Reliability of a performance estimate is defined as how closely a validation design's estimate (of AUC, Brier score and calibration) matches the model's realised performance on later, untouched data in the primary deployment setting.

3. Objectives

No.	Objective
O1	**Define the prediction task and audit the data.** For each dataset, fix the decision time, unit of analysis, rules for when an outcome is available, the eligible population, and which predictors are available at decision time. Specify how cancellations, incomplete outcomes and returning versus new customers are handled.
O2	**Quantify sensitivity to the validation design.** Compare random row-level, order-grouped, customer-grouped, chronological and rolling-origin evaluation across three model families and the baselines. Measure discrimination, calibration and model rankings, with controlled training size, a fixed tuning budget and confidence intervals that respect dependence.
O3	**Identify mechanisms.** Use fixed-test-set experiments that vary the training set's group overlap and time coverage one at a time, and simulations with known dependence and temporal change, to estimate the separate and interacting effects of each.
O4	**Examine reliability over time and changes in the service promise.** Test whether shifts in features, in lateness prevalence and in promised delivery windows are associated with later deterioration in discrimination and calibration. Analyse actual fulfilment time, promised time and delay relative to the promise separately.

No.	Objective
O5	**Produce a reporting protocol.** Turn the findings into a reproducible protocol for validating and reporting supply chain prediction models, including the minimum evidence needed to support a performance claim.

4. Research questions and hypotheses

All hypotheses, margins and tests are registered before the final evaluation on held-out data. Hypotheses state expected directions; they do not predetermine findings.

Research question	Hypothesis	Test
RQ1. How sensitive are performance estimates to the validation design?	H1. In DataCo, row-level random AUC exceeds order-grouped AUC by more than the equivalence margin δ. H2. In Olist, chronological AUC is lower than order-level random AUC by more than δ.	Paired cluster-bootstrap interval for the difference, compared with δ
RQ2. Do design contrasts identify the mechanism?	H3. In the DataCo overlap experiment, AUC increases monotonically with the share of test orders whose sibling items appear in training. H4. In simulation, the random-vs-grouped contrast tracks the injected dependence, and the grouped-vs-chronological contrast tracks the injected temporal change, with an interaction when both are present.	Slope test; recovery error against known truth
RQ3. Which changes over time are associated with deterioration?	H5. Feature drift (PSI) is weakly associated with the next period's change in AUC, whereas changes in lateness prevalence are associated with calibration-in-the-large error.	Origin-level regression with interval estimates
RQ4. Do validation choices change model selection?	H6. Model rankings under row-level random validation differ from rankings under the primary design.	Kendall's τ with bootstrap interval; rank-change frequency

Equivalence margin: *δ = 0.01 AUC, set in advance as the smallest difference that would plausibly change a model choice. Results are reported as superior, inferior, equivalent or inconclusive (Lakens, 2017). Multiple hypotheses are adjusted with Holm's method.*

5. Prediction tasks and populations

5.1 Estimands

Evaluation setting	Question answered	Role
Chronological, orders intact, returning customers allowed	How well does the model predict future orders in the existing customer population?	Primary
Chronological, new customers evaluated separately	How well does it transfer to future customers it has not seen?	Secondary
Customer-grouped random	How well does it predict for customers it has not seen, ignoring time?	Comparison
Order-grouped random	How well does it predict unseen orders, ignoring time?	Comparison
Row-level random	The common practice in published studies	Comparison

A returning customer in both training and test data is not treated as leakage when the task is to predict future orders, because their history is legitimately available at decision time. In DataCo a row is an order line item; in Olist a row is an order, so row-level and order-level random splits coincide for Olist.

5.2 Populations, outcomes and predictors

	DataCo	Olist
Decision time	When the order is placed	At purchase
Primary population	Non-cancelled orders with an outcome that can be evaluated (95.7% of rows); January 2015 – September 2017. Sparse single-market months (October 2017 – January 2018) are used only in a sensitivity analysis.	Delivered orders purchased January 2017 – August 2018 whose promised delivery date falls at least two weeks before the last recorded delivery, so that late outcomes can be observed
Outcome	Late_delivery_risk	Delivered after the promised date
Excluded predictors (known only after the decision)	Actual shipping days, delivery status, order status, shipping date; profit fields in the primary analysis, tested in sensitivity	Approval, carrier and delivery timestamps; review data
Outcome-availability risk	Cancelled orders are coded "not late" in the source	About 1–2% of July–August 2018 orders are undelivered at the data cut-off; slow orders are under-represented
Sensitivity analyses	Cancellations included as coded; cancellation as a separate outcome; profit fields added	Undelivered orders past their promise treated as late; alternative cut-off dates
Baselines	Shipping-mode only; scheduled-days only	Promised-delivery-time only; customer-region only

6. Data audit: preliminary signals

These single-run observations motivate the hypotheses. They are not findings and will be re-estimated under the full design.

- **DataCo cancellations.** All 7,754 cancelled rows (4.3%) are coded as not late. Among non-cancelled rows, First Class orders are late in 100% of cases. Shipping mode is a strong structural predictor, which is why it is used as a baseline.
- **DataCo grouping.** Each order has 2.75 line items on average, and every item in an order has the same label. A preliminary row-level random split gave an AUC of 0.821, against 0.760 for an order-grouped split and 0.762 for a chronological split. This suggests that DataCo estimates are more sensitive to keeping orders intact than to time.
- **Olist over time.** A preliminary random split gave 0.784, against 0.647 for a chronological split, suggesting greater sensitivity to temporal evaluation. The monthly late rate ranged from about 1% to 21%.
- **Olist service promise.** The lateness spikes in November 2017 and February–March 2018 coincided with slower fulfilment (median 12.7–14.3 days, against about 10) while promises stayed stable. The August 2018 rise coincided with a shorter promise (median 13.4 days, against about 22) while fulfilment became faster.

7. Research design

7.1 Comparing validation designs (O2)

- **Models:** regularised logistic regression, random forest and gradient boosting, plus the baselines in Section 5.2.
- **Equal footing:** training size is matched in unique orders as well as rows; every model gets the same tuning budget, tuned by time-aware inner validation inside the training data; all preprocessing, encoding and calibration are fitted within each training set.
- **Metrics:** ROC AUC and average precision for discrimination; Brier score, calibration-in-the-large, calibration slope and expected calibration error for calibration (Van Calster et al., 2019); Kendall's τ

for model rankings.

- **What is reported:** differences between designs are reported as **validation-design contrasts**, not as causes. Each comparison reports the number of unique orders, training duration, test horizon and training recency.

7.2 Fixed-test-set mechanism experiments (O3)

The test set is held fixed (all orders in a designated evaluation period) and only the training set is changed, with its size held constant. This isolates one factor at a time:

Training set	Group overlap with the test set	Time coverage	Isolates
A. Past only	None	Before the test period	Reference (primary setting)
B. Past and contemporaneous, test orders excluded	None	Includes the test period	Effect of time coverage
C. As B, plus sibling items of a controlled share of test orders	0% to 100% in steps	Includes the test period	Effect of order dependence
D. Past only, plus earlier orders from the test customers	Customer-level only	Before the test period	Legitimate customer history

Comparing B with A estimates the effect of time coverage at zero overlap. The dose-response across C estimates the effect of order dependence (H3). The design is repeated across several evaluation periods.

7.3 Simulation with known mechanisms (O3)

Semi-synthetic outcomes are generated on the real predictors, with known levels of within-order correlation, covariate shift, base-rate shift and promise change, applied singly and in combination in a factorial design. Recovery is measured as the error between the estimated contrasts and the effects that were built in (H4).

7.4 Reliability over time (O4)

Rolling-origin evaluation is analysed at the level of each forecast origin. At each origin the study records feature drift (PSI and standardised mean shift), the change in lateness prevalence, the change in the promised window, the change in fulfilment time, and the following period's changes in AUC and calibration. For Olist, an auxiliary model predicting delivery duration separates operational change from changes in the promise. Associations are reported as associations (H5), with no causal claims.

7.5 Uncertainty that respects dependence

- Paired bootstrap on the shared test set, resampling whole orders, and customers where the estimand requires it (Field & Welsh, 2007).
- Resampling of weekly blocks, or inference at the level of each origin, for chronological comparisons (Künsch, 1989).
- Intervals are estimated for each difference directly. Repeated random splits are not treated as independent replications.

7.6 Safeguards against tuning to the results

- **Development:** the analysis code, defaults and thresholds are developed on DataCo 2015–16 and on the simulations only.
- **Final test:** the final evaluation periods for both datasets, and all of Olist's confirmatory results, stay untouched until the analysis plan is registered.
- **Robustness:** results are reported for every model family and both datasets, plus the sensitivity analyses in Section 5.2.

7.7 Software

The analysis uses Python and scikit-learn. The BDP Model Gate (version 0.6.0) produces standardised validation, calibration and drift reports for every design, and its outputs are reported descriptively. Whether the gate supports better release decisions is evaluated separately in Paper B. Code, configuration and the registered analysis plan will be released openly.

8. Expected contributions

- **Methodological:** a framework for quantifying how sensitive performance estimates are to the validation design. It separates descriptive contrasts from mechanisms identified with fixed-test-set experiments and simulations, and uses uncertainty methods suited to grouped and time-ordered data.
- **Empirical:** evidence across two datasets on when order dependence, changes in the service promise and temporal shift alter performance, calibration and model rankings, including when distribution change occurs without material deterioration.
- **Practical:** a reporting protocol stating the minimum evidence needed before a performance claim for a supply chain model can be relied on.

9. Target journals

International Journal of Production Economics; *International Journal of Production Research*; *Decision Support Systems*; *International Journal of Forecasting*. Check current quartile rankings before submission.

10. Scope and limitations

- Both datasets concern late-delivery prediction. Generalising to other tasks, such as demand forecasting, is left to future work.
- DataCo's outcome appears partly rule-based, which limits external validity. This is why the study uses a second dataset and simulations.
- Conclusions apply to orders whose outcome can be evaluated at the stated decision time.
- The evaluation of automated governance checks and release decisions is deliberately excluded and forms Paper B.

References

Bergmeir, C., & Benítez, J. M. (2012). On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191, 192–213.

Constante, F., Silva, F., & Pereira, A. (2019). *DataCo smart supply chain for big data analysis* [Data set]. Mendeley Data.

Eck, B., Kabakci-Zorlu, D., Chen, Y., Savard, F., & Bao, X. (2022). *A monitoring framework for deployed machine learning models with supply chain examples*. arXiv:2211.06239.

Field, C. A., & Welsh, A. H. (2007). Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B*, 69(3), 369–390.

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804.

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 15.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241.

Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4), 355–362.

Oden, V. (2026). *bdp-model-gate* (Version 0.6.0) [Software]. Python Package Index.

Olist. (2018). *Brazilian e-commerce public dataset by Olist* [Data set]. Kaggle.

Roberts, D. R., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929.

Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4), 437–450.

Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., & Steyerberg, E. W. (2019). Calibration: The Achilles heel of predictive analytics. *BMC Medicine*, 17, 230.